



AL FLASHER 12 V OPERATING VOLTAGE TEST REPORT

DATE: 17.04.2026

PAGE NO.: 1/3

Part Name	AL FLASHER 12 V	Application	COMMERCIAL APPLICATION
Customer Part No.	FGG00600	Customer	ASHOK LEYLAND
UCAL Part No.	RDEPL/AL/15	Report No	R&D/AL/12VF/VAL/26/011

INTRODUCTION:

UCAL is currently supplying 12V AL Flashers to Ashok Leyland for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 12 V flashers assembly after **Operating voltage test**.

TEST SPECIFICATION:

Test standard as per IS 13135. Sample to be put in to testing with a voltage 12 V.

ACCEPTANCE CRITERIA:

Class A: All function of the flasher performs as designed before and after the test.

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:02	O	O	O

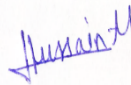
O: OK

X: NG

Δ: To be confirmed

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

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Prepared By	Approved By



AL FLASHER 12 V OPERATING VOLTAGE TEST REPORT

DATE: 17.04.2026

PAGE NO.: 2/3

PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	8 V~16 V
2	Test voltage	13.5 V
3	Flash rate(Turn signal/Hazard warning)	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST PROCEDURE:

- The DUT unit shall be mounted as in vehicle condition applied voltage of 13.5V for 1 hour.
- Lamp loads shall be connected across the output and ground,
- Direction indicator: 2 X 21W + 1X1W+2W CLUSTER SIGNAL
- Hazard :4 X 21W + 2 X1W+2 X 2W CLUSTER SIGNAL

TEST SETUP:



TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL



AL FLASHER 12 V OPERATING VOLTAGE TEST REPORT

DATE: 17.04.2026

PAGE NO.: 3/3

Test Observation during Test:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:01	
Turn Signal RH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	85	OK
Turn Signal LH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	85	OK
Hazard Mode	2x (2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal)	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	170	OK
Duty cycles for RH&LH	2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal	49%	OK
Duty cycles for Hazard Mode	2x (2 x 21 W Lamp + 1x 2 W LED+ 2 W Cluster Signal)	49%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:02	AFTER TEST SAMPLE – S:02
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test (Class A).